

# Aegis Hackathon Platform

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An online hackathon workspace where participants join events, form teams, tackle daily challenges, submit work, and climb the leaderboard — while admins manage the full event lifecycle from a single dashboard.

Built as a progressive web app with offline support, so participants stay productive even when connectivity drops.

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## Users

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| Role           | Capabilities                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------------|
| Public visitor | View event info, public leaderboard, sign-in page                                                         |
| Participant    | Browse challenges, schedule, and rules; join or create teams; submit work; track progress; manage profile |
| Team lead      | Approve or reject join requests; manage team roster before lock                                           |
| Admin          | Configure hackathons, manage all content, review and score submissions, monitor progress                  |

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## How It Works

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### Active Hackathon

The platform presents **one active hackathon** to participants at a time. Each hackathon bundles its challenges, schedule, rules, skill tracks, and policies (team size, late-submission handling) into a single event.

### Event Phases

Every hackathon moves through three phases — **upcoming**, **live**, **completed** — and the homepage always reflects the current state.

## Online-First Delivery

Schedule items reference streams, chat channels, mentor rooms, and portals. No physical venue is required.

## Teams

Participants create a team or request to join one. Team leads approve requests. Once a team records its first submission, the roster **locks** — no more joins or departures. One team per participant per hackathon.

## Challenges

Daily challenges unlock on a schedule. Each has a brief, resources, point value, and a deadline. Late submissions are accepted with the hackathon's configured penalty.

## Submissions

Participants submit a project title, description, category, and delivery links against a challenge. Admins review and assign a raw score; the platform applies any late penalty to produce the final score used on leaderboards.

## Skill Tracks

Optional learning modules participants can complete alongside challenges.

## User Profile

Participants have a profile with display name, avatar (URL-based), and bio. The avatar appears in the navbar and sidebar. Email is managed by the auth provider and displayed read-only.

## Offline Support

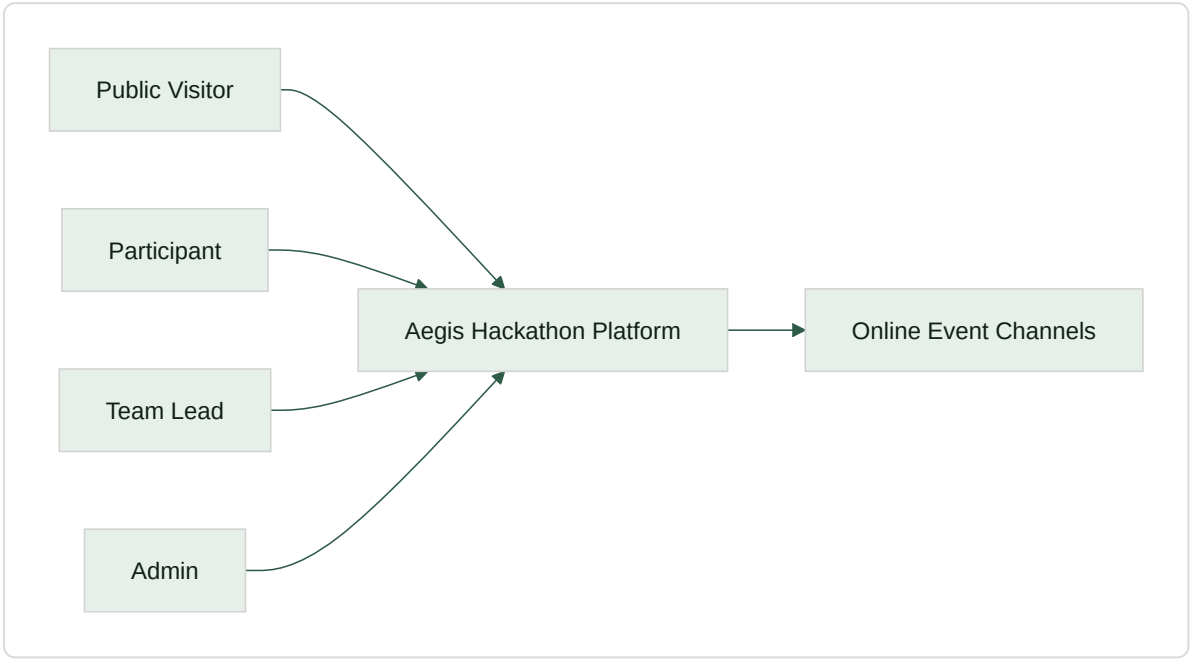
Previously loaded pages remain accessible during connectivity loss. Submissions queue locally and sync automatically when the connection returns.

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# Diagrams

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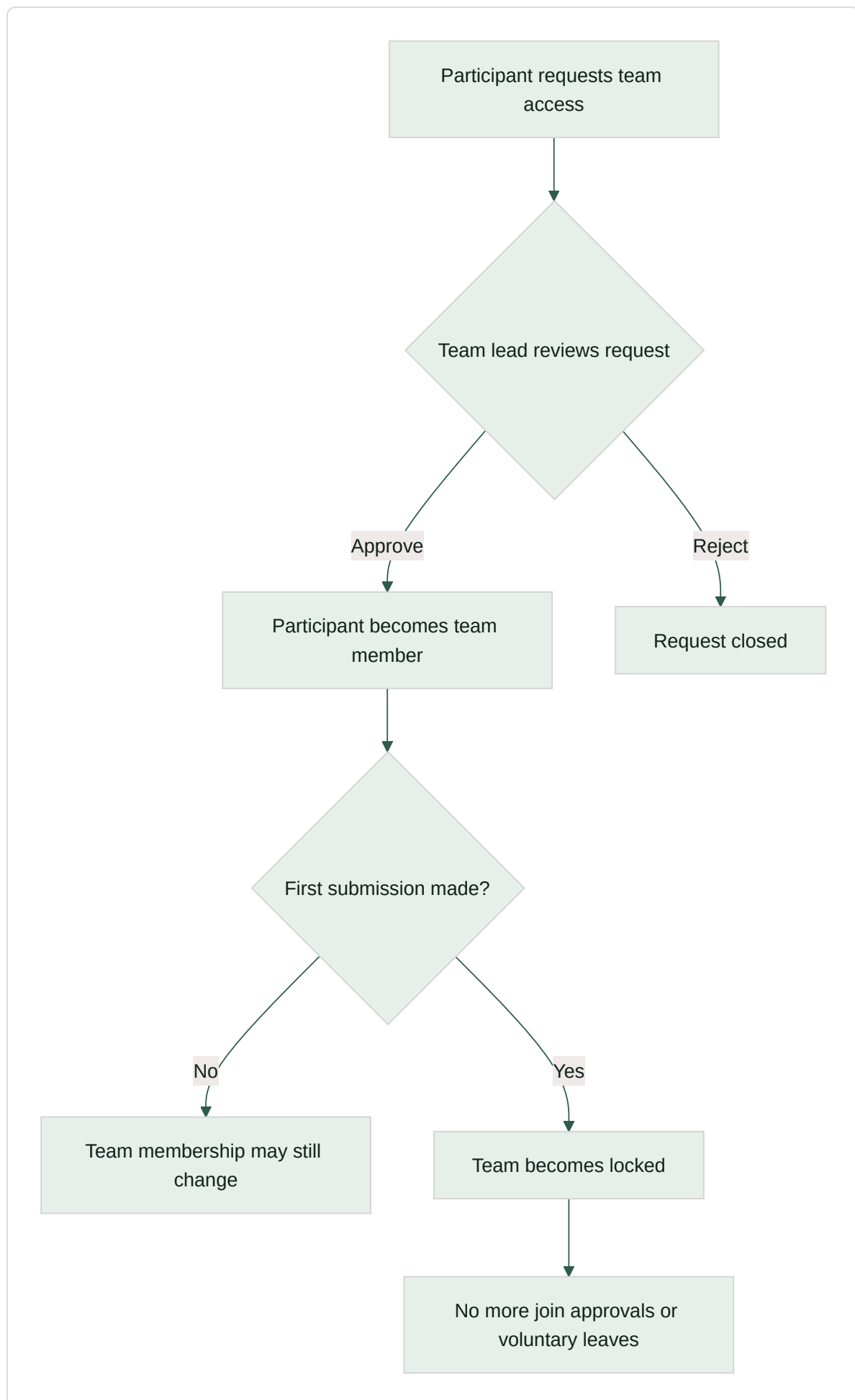
## Context



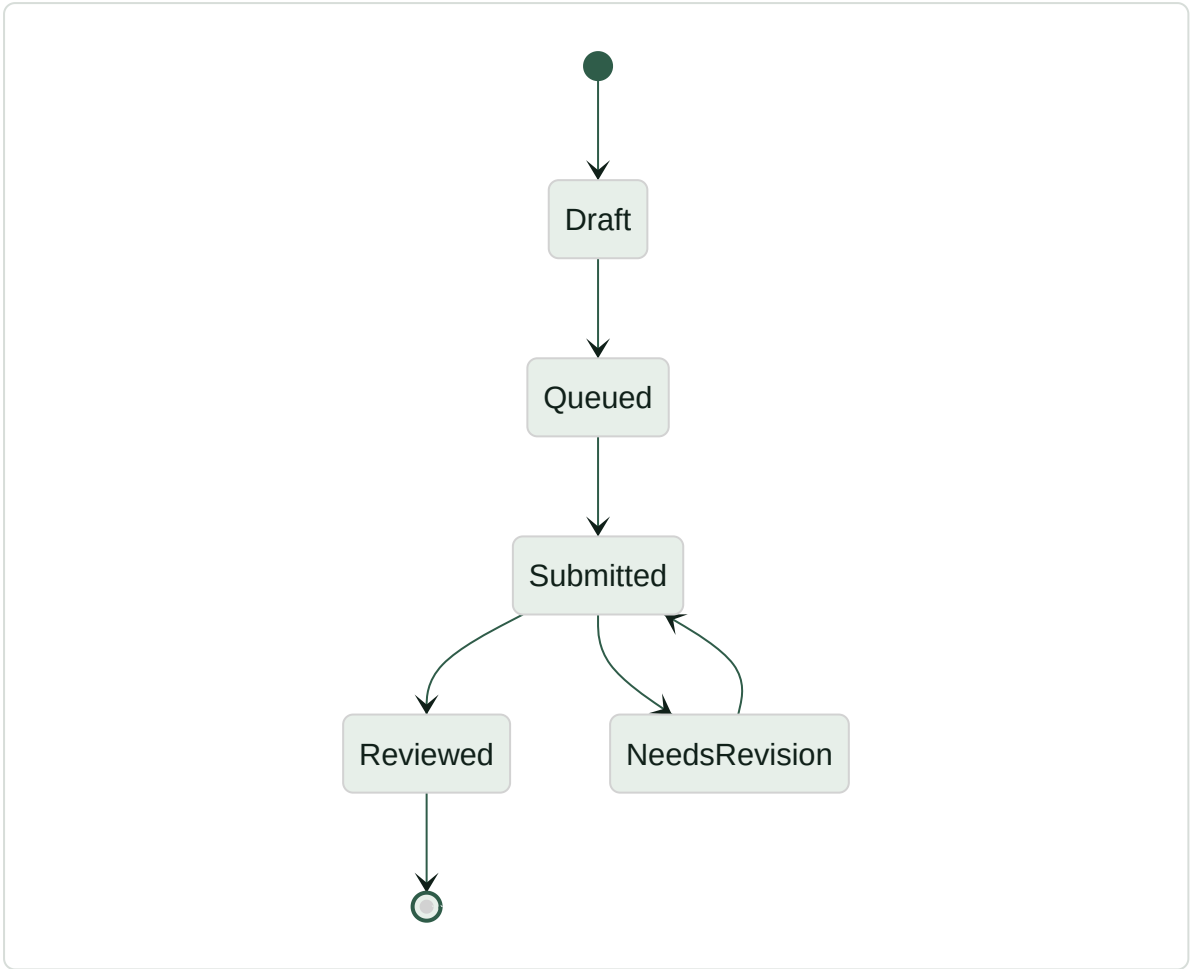
## Participant Flow



**Team Access Control**



## Submission Lifecycle



## Pages

| Area        | Pages                                                                                                                                |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Public      | Landing, Login, Public Leaderboard                                                                                                   |
| Participant | Dashboard, Challenges, Challenge Detail, Leaderboard, Teams, Rules, Skill Tracks, Submit, My Submissions, Submission Detail, Profile |
| Admin       | Overview, Hackathons (full CRUD), Submissions (review/score), Progress                                                               |

Empty states show contextual guidance instead of blank content.

# Business Rules

- One active hackathon at a time
- Unique team names within a hackathon
- One team per participant per hackathon
- Team join requires lead approval
- Teams lock after first submission
- Challenge access follows published unlock times
- Late submissions follow the hackathon's configured penalty policy
- Admins manage all content through the UI

# Tech Stack

| Layer    | Technology                                                |
|----------|-----------------------------------------------------------|
| Frontend | React 19, Vite 7, Zustand, TypeScript                     |
| Backend  | Express 5, Node 20, TypeScript                            |
| Database | PostgreSQL (Neon) with Drizzle ORM                        |
| Auth     | Neon Auth (wraps better-auth), JWT with JWKS verification |
| Offline  | IndexedDB via idb, sync queue, Workbox service worker     |
| Styling  | Vanilla CSS with design tokens                            |
| Monorepo | pnpm 10 workspaces (client/, server/)                     |

# Hosting & Deployment

| Service | Purpose                    | Tier |
|---------|----------------------------|------|
| Render  | App hosting (Docker)       | Free |
| Neon    | PostgreSQL database + auth | Free |

The repo includes a `render.yaml` blueprint for one-click Render deployment. Connect the GitHub repo, set the secret env vars, and deploy. The Dockerfile builds a production image serving both the API and static frontend on port 8080.

A `fly.toml` is also included as an alternative for Fly.io.



## Environment Variables

| Variable      | Description                         |
|---------------|-------------------------------------|
| DATABASE_URL  | Neon PostgreSQL connection string   |
| NEON_AUTH_URL | Neon Auth endpoint URL              |
| CORS_ORIGIN   | Allowed origin(s) for CORS          |
| NODE_ENV      | production in deployed environments |
| PORT          | Server port (default 8080)          |

## Non-Functional Requirements

- **Offline resilience** — Core pages and queued submissions survive connectivity loss
- **Security** — JWT-authenticated routes; admin-only endpoints; participants cannot access others' private data
- **Usability** — Current event state, deadlines, and next actions are always visible; empty states provide guidance
- **Maintainability** — This document is updated when user-visible behavior changes

## Known Limitations

- Binary media upload is not yet exposed in the submission flow
- Offline support covers navigation and queued submissions, not a full local data replica